ENTERPRISE MARKET RISK

TOWARD THE NEW NORMAL

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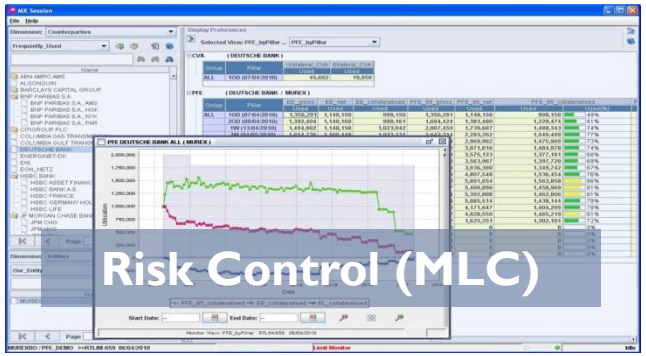
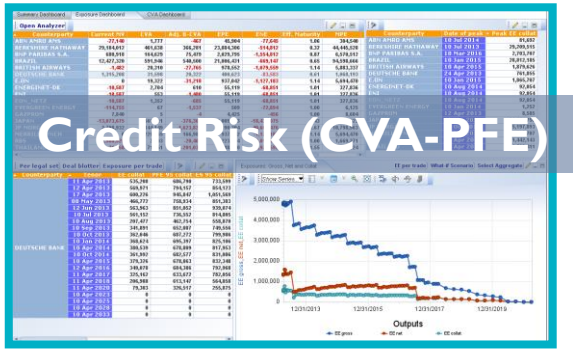
Murex Solutions for Risk Management





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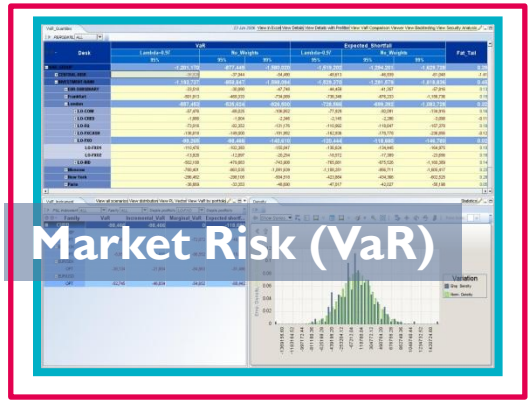

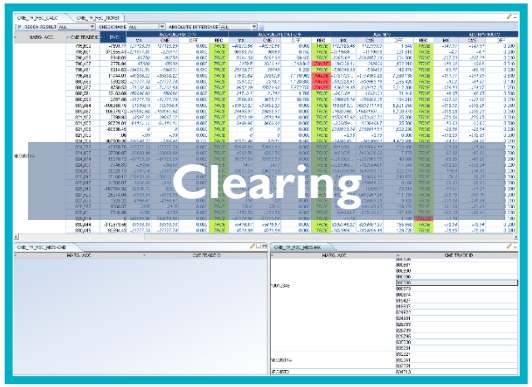

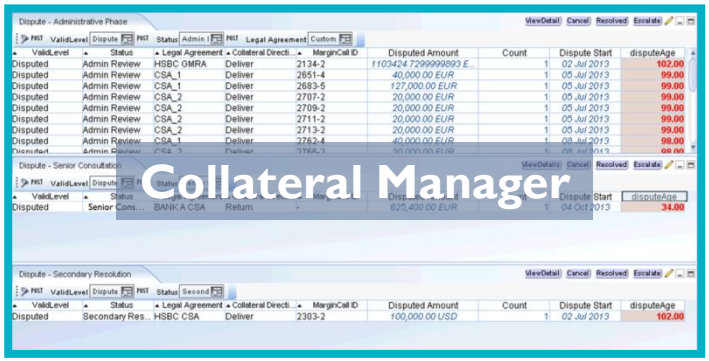

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VaR Introduction

A	Context : Basel regulation
B	VaR definition
C	VaR computation process
D	Backtesting

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Objectives

By the end of this session you will be able to answer :

- **What does a VaR figure represent?**
- **What is it used for?**

- **What are the hypotheses of VaR?**
- **What are the main steps of a VaR calculation?**
- **What is backtesting?**

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VaR Introduction

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B		VaR definition
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	1	Scenario generation
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	3	Worst scenario selection
	4	Various methodologies
D		Backtesting
E		Conclusion

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VaR in Basel context

		BANKING	TRADING
CREDIT RISK	Standard		
	Model		
MARKET RISK	Standard		
	Model		

Where does VaR stands in this matrix?

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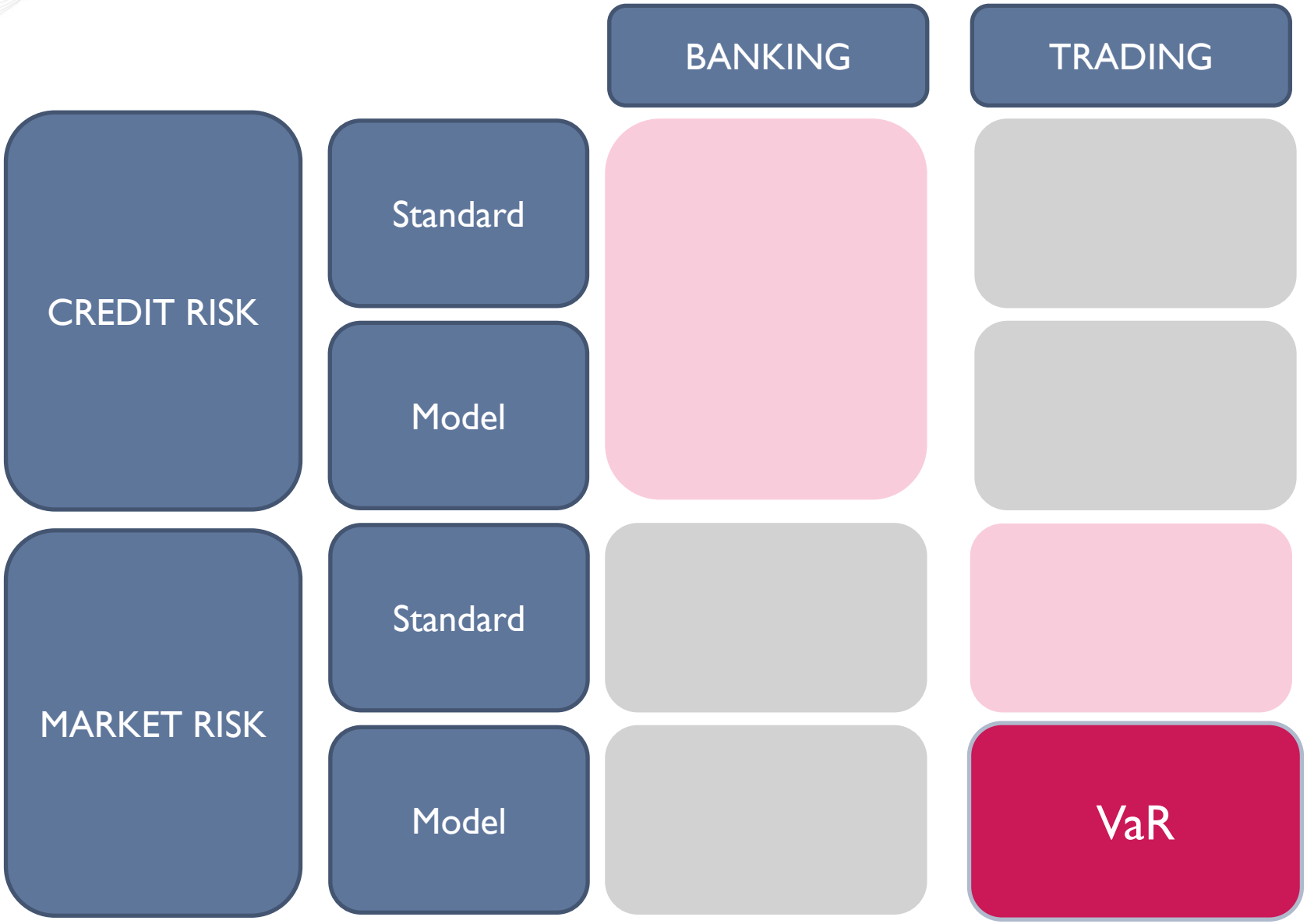
VaR in Basel context



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VaR definition

How would you define VaR?

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VaR definition

The Value at Risk measures the POTENTIAL LOSS in value of a risky asset or portfolio

over a defined PERIOD OF TIME

for a given CONFIDENCE INTERVAL.

Can you rephrase : 10 Day VaR 95% = 10M ?

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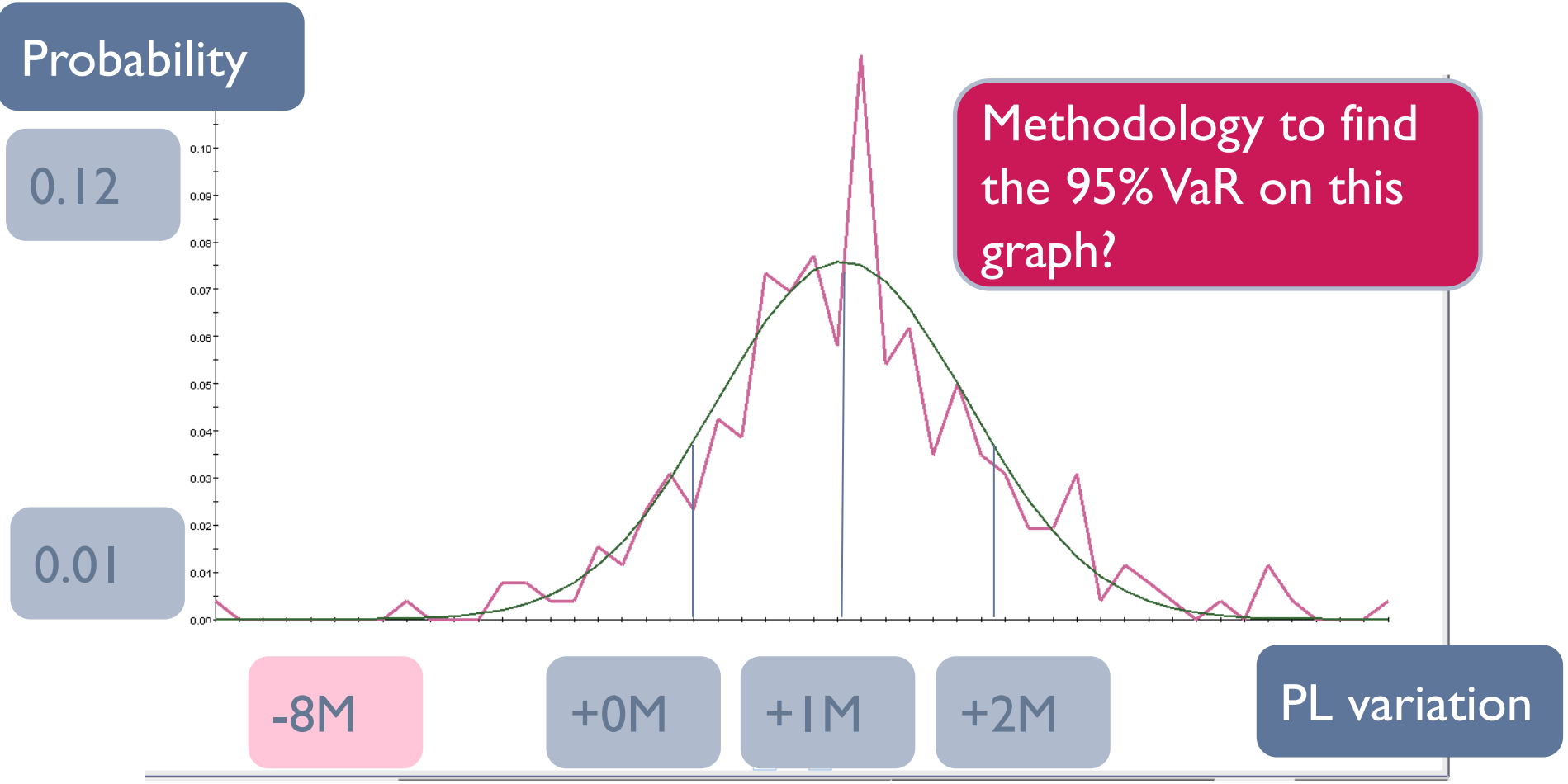
Distribution

Distribution of Potential 1day PL variations



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VaR definition

What is VaR figure used for?

The Value at Risk usage:

- Internal use
- External communication (Pilar 3)
- Capital requirement (> 3*VaR)

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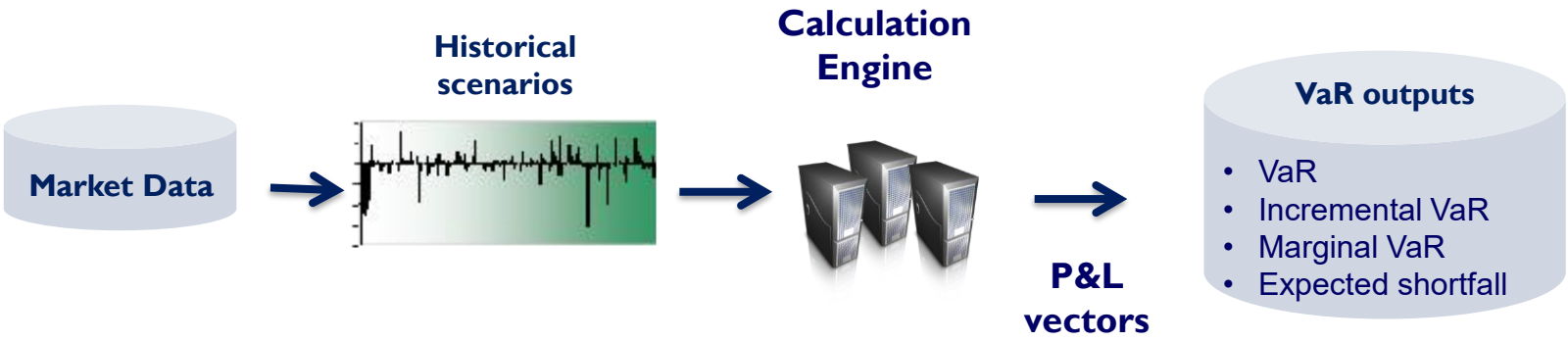
HVaR Full revaluation in 3 Steps

The Historical VaR methodology consists of 3 phases:

1. Scenarios Generation
2. Revaluation of positions based on the shifted market parameters
3. Reporting of aggregated results



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Scenario generation

To transform historical prices into shifts:

Standard

$\delta_i = S_i -$

Percent

Log

Can you compute shifts on given historic?

Scenario Start Date	Scenari...	Main L...	Second...	Initial Value
Sep 09 2011	Security ...	BAS.DE	DE XET...	31.50
Sep 12 2011	Security ...	BAS.DE	DE XET...	32.00
Sep 13 2011	Security ...	BAS.DE	DE XET...	33.48
Sep 14 2011	Security ...	BAS.DE	DE XET...	34.00
Sep 15 2011	Security ...	BAS.DE	DE XET...	30.00
Sep 16 2011	Security ...	BAS.DE	DE XET...	32.00

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Scenario generation

A scenario is a set of shifts on market data.

Scenario Number	Scenario Type	Main Label 0	Second Lab...	Nb Da...	Variation
1	FX Spot	USD	EUR	0	0.0000000
	Security Spot	BAS.DE	DE XETRA	0	0.5000000
	Zero coupons	EUR :STD	EUR	91	0.0436332
				366	0.0404528
				3,653	0.0018824
2	FX Spot	USD	EUR	0	0.0000000
	Security Spot	BAS.DE	DE XETRA	0	1.4800000
	Zero coupons	EUR :STD	EUR	91	0.0000000
				366	0.0158021
3	FX Spot				
	Security Sp				
	Zero coup				
4	FX Spot				
	Security Spo				
	Zero coupons	EUR :STD	EUR	91	0.1137177
				366	-0.2384276
				3,653	-0.0772488

How many shifts are needed to impact:

- an equity price
- a credit curve
- a FX smile curve

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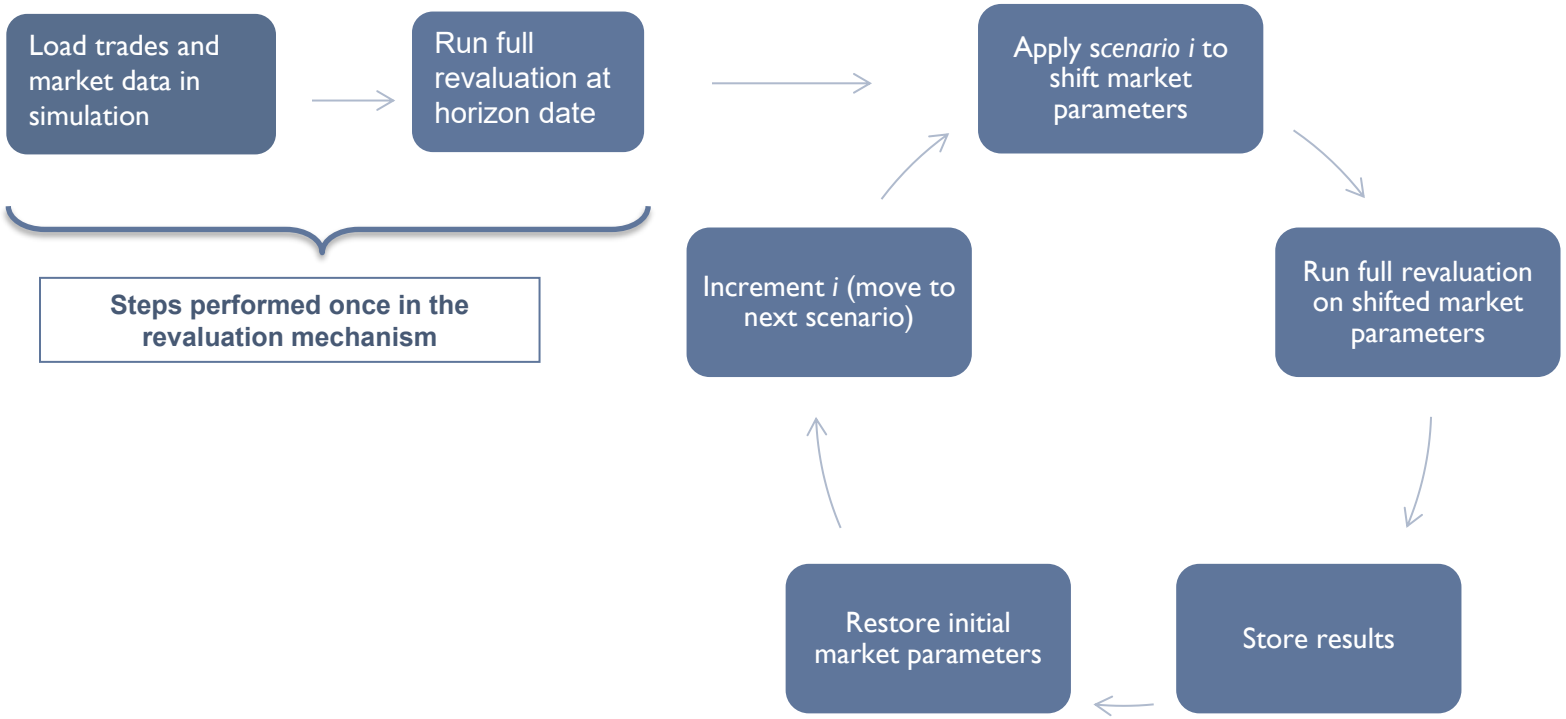
PL vector computation

PL vector computation is based on MX simulation engine.

Positions are revaluated by the **sequential** application of scenarios.



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Part 1 : Var Introduction

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Aggregation

From P&L vectors (ie. one line per scenario and position) to VaR results (ie. worst case for each position)

Scenarios	P&L variation
SCN1	-100
SCN2	+145
SCN3	+130
SCN4	-125
SCN5	-98
SCN6	+128
SCN7	-149
SCN8	+99
SCN9	-145
SCN10	-113
SCN11	+150
SCN12	+86
SCN13	-105
⋮	
SCN251	+108
SCN252	-119

→

VaR Scenarios	P&L variation sorted asc.
SCN7	-149
SCN9	-145
SCN4	**-125**
SCN252	-119
SCN10	-113
SCN13	-105
SCN1	-100
SCN5	-98
SCN12	86
SCN8	99
SCN251	108
SCN6	128
SCN3	130
⋮	
SCN2	145
SCN11	150

In the illustrated example, if we suppose we have **252 scenarios** defined and **99%** confidence level; the VaR will thus be the **3rd worst P&L** (i.e. 252*0.01)

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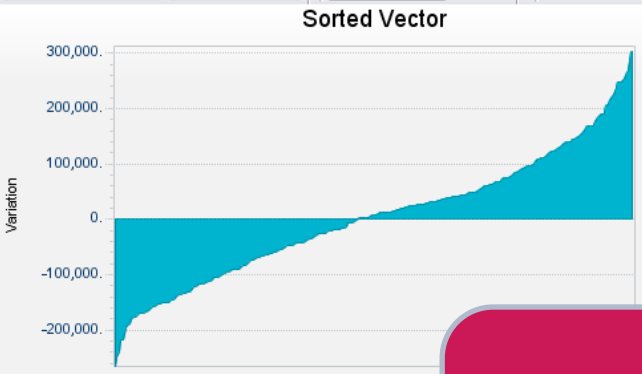
Aggregation

Distribution | Vectors | Graph | Maps | Trends | RiskFactors | Greeks | What-if

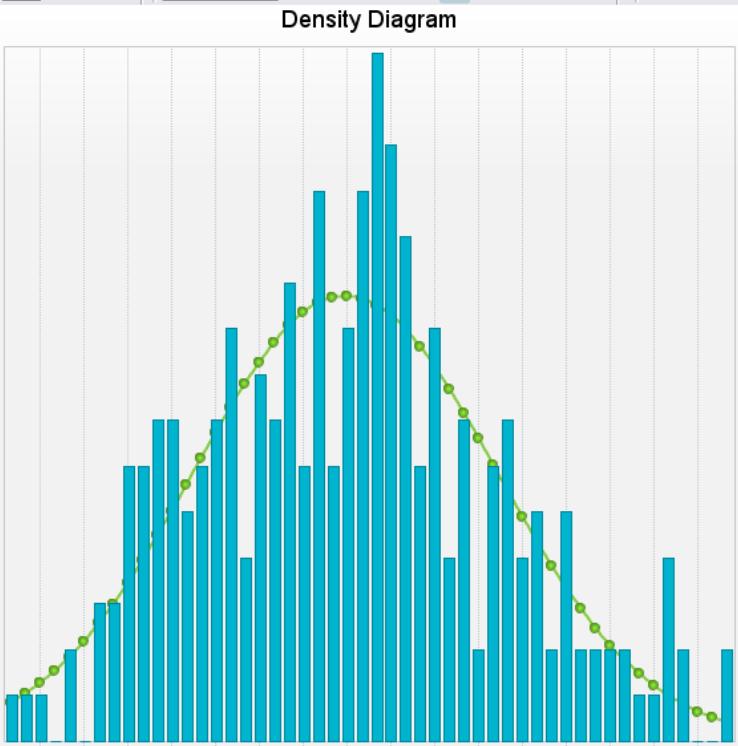
Global	Desk	SubDesk	Reference VaR	VaR	EVT VaR	EWM
US	Total US		-464,339,940	-464,339,940	-451,136,643	-31
	CANADA EQ	CA EQ TRADING	-2,177,153	-2,177,153	-2,235,070	-2
	CANADA FI	CA FI BOND	-720,062	-720,062	-705,236	
	CANADA FX	CA FXO	-5,251,998	-5,251,998	-5,264,287	-5
	CANADA IRD	CA IRD	-169,428	-169,428	-171,797	
	USA CR	Total USA CR	-883,198	-883,198	-851,822	
		US CR FLOW	-81,866	-81,866	-82,932	
		US CR STRUCT	-866,196	-866,196	-857,020	
	USA EQ	Total USA EQ	-450,397,080	-450,397,080	-449,762,007	-31
		US EQ CONV	-438,425,720	-438,425,720	-448,803,536	-32
		US EQ EXO	-1	-1	-1	
		US EQ HEDGE	-714,349	-714,349	-709,050	
		US EQ SECFIN	-488,050	-488,050	-472,013	
		US EQ TRADING	-47,132,943	-47,132,943	-42,738,676	-4
	USA FI	US FI BOND	-27,719,379	-27,719,379	-27,050,746	-32
	USA FX	Total USA FX	-1,127,767	-1,127,767	-1,139,648	-2
		US FX	-1,003,265	-1,003,265	-974,639	-1

Variations | Market Risk Views | Virtual Sources | Show serie...

	Result
2	-265,441.
122	-248,603.
107	-237,903.
68	-217,179.
56	-216,364.
140	-197,039.
117	-190,714.
95	-188,654.
32	-178,670.
22	-177,295.
171	-176,911.
10	-170,933.
75	-169,428.
161	-168,656.
9	-168,455.
221	-166,354.



0.00 | 100.00 | 50.00 | Show serie... | 3D



Is the VaR on the book equal to the sum of VaRs on each instrument?

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VaR scenario-based methodologies

Scenario generation	- Historical data - Monte Carlo generation
P&L computation	- Full revaluation - Taylor approximation

Pros and cons of each methodology?

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Backtesting definition

"*VaR is only as good as its backtest. When someone shows me a VaR number, I don't ask how it is computed, I ask to see the backtest.*" (Brown, 2008)

Backtesting is a statistical procedure where actual profits and losses are systematically compared to corresponding VaR estimates.

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Backtesting definition

Date	One-day 99% VaR predicted	Daily P&L observed
D1	-1M	+0.2M
D2	-1.1M	-1.2M
D3	-0.9M	-0.3M
…	…	…
D250	-1.2M	+0.5M

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Risk manager job

On his dashboard, the risk manager will have to explain:

- Var figure on a given desk, if its value has increased between yesterday and today
- P&L loss between yesterday and today if it does exceed yesterday published var (what we call a backtesting break)

Level 1	Today VaR	Yesterday VaR	VaR Ratio	Today scenario	Yesterday scen...	P&L	PL to VaR
F. CURRENCIES	-355,049	-153,017		May 31 2011	21 Apr 2011	62,344	
USA FI	-27,719,379	-53,311,842		May 13 2011	18 May 2011	1,869,357	
US FI BOND	-27,719,379	-53,311,842		May 13 2011	18 May 2011	1,869,357	
USD_TSY_BD	-295,927	-290,572		Aug 11 2010	18 Jul 2011	-356,540	
TD-CL-FI-CLTR	-30,842	-0		Aug 17 2010	13 Jul 2011	0	
NE-CL-FI-CLTK	-3,499,132	-6,252,442		Aug 20 2010	29 Mar 2011	316,599	
CIBC-HL-FI	-55,351	-0		Sep 10 2010	13 Jul 2011	0	
GS-HL-FI	-2,289,865	-5,415,054		Oct 18 2010	12 May 2011	291,579	
SCO-CL-FI-CLTO	-79,499	-56,898		Oct 19 2010	09 Feb 2011	4,195	
JPM-HL-FI	-112,740	-7,490,112		Oct 21 2010	07 Mar 2011	-302,996	
JPM-CL-FI-CLTG	-48,455	-3,537,088		Nov 15 2010	12 May 2011	-228,433	

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ANY QUESTIONS?

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THANK YOU

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